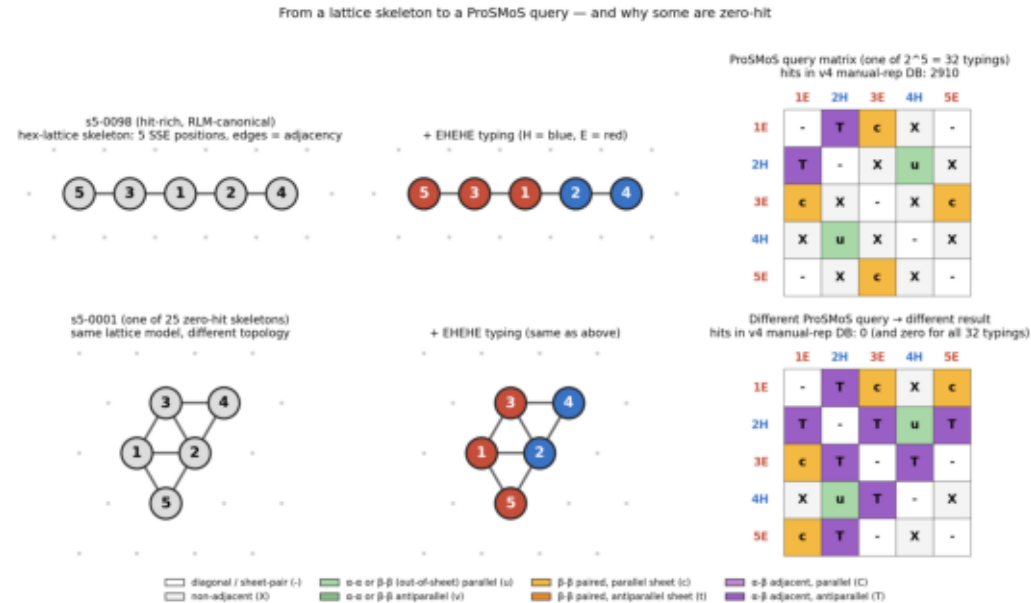


Enumerated S5 negative space

What the 2D-hex-lattice SSP model can make — and what nature does not

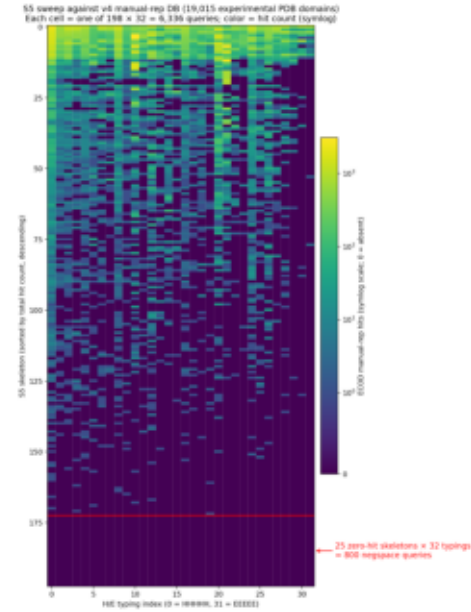
We enumerated every 5-SSE arrangement the Chitturi 2016 lattice permits (198 skeletons × 32 H/E typings = 6,336 queries), asked whether each occurs as a substructure of any protein, and found 25 skeletons — 800 queries — absent from 5.4 million real and predicted domains. Along the way we found and fixed four latent bugs in ProSMoS searchmatrix.

The unit of work: a skeleton + a typing = a ProSMoS query



- Skeleton: 5 SSE positions on the 2D hex lattice + adjacency + per-triple handedness. 198 unique S5 skeletons after SCC-2 and compactness filters.
- Typing: assign H or E to each of the 5 positions. $2^5 = 32$ typings per skeleton.
- ProSMoS query: skeleton $\times$ typing $\rightarrow$ interaction matrix with codes for parallel/antiparallel β -pairing, α - α packing, mixed α - β contact.

The empirical landscape: 198 × 32 hits against 19k curated P



- Cells = individual queries (6,336). Sorted by skeleton total hits, descending.
- 25 skeletons × 32 typings = 800 negspace queries — the black block at the bottom.
- Composition gradient visible as vertical bands (strand-rich typings darker).
- Bright vertical stripes at typings 10 & 21: EHEHE Rossmann-type sheets.

The question — is the gap real or an artifact of curation?

- 19k manual reps is ~4% of experimental PDB-derived ECOD domains, and a vanishing fraction of sequenced protein space.
- Hypothesis A — the 25 zero-hit skeletons are just under-sampled by the curated set; they light up when we search broader corpora.
- Hypothesis B — the gap is real: these motifs are absent from nature (or absent from observable nature — a subtly different thing).

The test: run the 800 negspace queries against progressively larger databases.

The result matters for what we can say. If (A), the enum's negspace is a manifesto of curation gaps. If (B), it's a manifesto of protein space.

Three databases, one 800-query corpus

- All three built via the same pipeline: PALSSE SSE assignment → generateMatrix → concatenated metamatricesDB.
- domain_summary.source_type='pdb' for the middle DB (not derived_files.domain_source_type — see next slide).

The result — 800 queries, 5.4M structures, zero hits

- truly_absent = 800, AFDB_only = 0, AFDB_loss = 0, common = 0
- Not a single motif from the enumerated negspace set appears as a substructure of any real or predicted protein at this scale.
- AFDB extending PDB by $\sim 10\times$ produced no new realizations. Either the motifs are unrealized, or AFDB inherits PDB's structural distribution from its training set. The data alone can't separate these two readings — but the practical consequence is the same: broader search does not fill the gap.

Getting here was hard: four latent bugs in searchmatrix

- ProSMoS searchmatrix is the reference tool for this kind of substructure search. The shipped 2010 binary and any binary rebuilt from unpatched source all hit at least one of four buffer / validation bugs in searchControl.h.
- Each bug produces the operator symptom "0 hits" — but for a different reason.
- Fixes at commits d3e1e83, 0d9e2b0, c2904f5. Canonical binary is c2904f5.
- The 800-query negspace claim was independently re-verified against each successive fix — it holds because the queries genuinely find zero matches, so the buffer bugs had nothing to drop and the validation bug caused a no-op that also (correctly) reported zero.

What's in flight now — the full 6,336-query landscape

- The 800-query negspace claim is the paper's headline; the next figure is the landscape it sits in.
- Full-S5 sweeps in progress (as of talk): PDB-exp ~55%, AFDB ~52%, both producing rich lit-up data (~1,400 and ~1,100 lit-up queries respectively).

- Deliverables:

- *Three-panel comparable hit grid: same skeleton ordering, three DBs, so the reader can visually scan across.*

- *Low-hit-skeleton spotlight: 5-10 skeletons that sit just above the negspace boundary (1-3 typings lit of 32). PyMOL renders of the actual ECOD domains that realized each rare typing.*

The 25-row negspace block should sit stably at the bottom of all three panels — the whole point of the figure is that the block doesn't move as DB scale grows.

The open question — geometric reachability

- The lattice enum produces skeletons satisfying planarity, SCC-2, compactness, per-triple handedness.
- It does not check:
- Resolution requires constructive 3D backbone modeling (RFdiffusion / Rosetta-style) on the 800.

- Steric overlap between SSEs (lattice nodes have zero volume; real helices/strands don't)

- Loop-length feasibility for connecting chain segments

- β -sheet H-bond register compatibility

- Ramachandran / SSE-packing statistical priors

Two competing readings of the 800-query zero:

1. Motifs are geometrically possible but unrealized by evolution — they are unexplored de novo design targets.

2. Motifs are geometrically impossible — the lattice abstraction over-generates relative to 3D space; the gap is a modelling artifact.

Recap

- Enumeration: 198 S5 skeletons $\times$ 32 H/E typings = 6,336 queries from a lattice model.
- Negspace identification: 25 of 198 skeletons return zero hits across every typing against ECOD's manual reps $\rightarrow$ 800 negspace queries.
- Scale-up test: 800 queries $\times$ (curated PDB + all experimental PDB + AFDB non-singleton) = 2,400 sweeps, 0 lit-up. 5.4M structures searched, no matches.
- Cost of getting there: four latent bugs in searchmatrix, all fixed in searchControl.h this session.
- Now in flight: full 6,336-query sweeps against PDB-exp and AFDB for the landscape figure + spotlight.
- Open: are the 800 sterically reachable but evolutionarily untaken, or lattice-model artifacts?